SUPERSTORE SALES DATA ANALYSIS & PERFORMANCE REPORT (2015–2018) USING PYTHON

Executive Summary

This report delivers a comprehensive data analysis of a nationwide retail superstore's performance between 2015 and 2018. Utilizing Python-based data science workflows, the study evaluates overall sales performance, profitability dynamics, geographic distribution, and product category efficiency. The primary goal is to surface actionable business intelligence to optimize pricing strategies, control discount erosion, and refine inventory logistics.

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1. Project Overview & Objectives

The primary objective of this project is to perform an end-to-end exploratory data analysis (EDA) on the Superstore sales dataset to evaluate core commercial metrics across product categories, geographic regions, customer segments, and temporal dimensions.

Key objectives include:

- **Revenue & Profitability Evaluation:** Identifying top-performing categories and diagnosing sub-categories driving margin dilution.
- **Geographic Distribution:** Mapping sales density and regional profit contributions to allocate operational resources effectively.
- **Temporal Trend Analysis:** Measuring Compound Annual Growth Rate (CAGR) and seasonal demand surges from 2015 through 2018.
- **Strategic Optimization:** Formulating data-backed recommendations regarding discount policies and category-specific management.

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2. Dataset Information & Attribute Dictionary

The analysis was executed on a structured transactional dataset comprising **9,994 records** and **21 attributes**, covering orders placed from 2015 to 2018.

Attribute Name	Data Type	Description
Order ID	Object	Unique identification code assigned to each transaction order.
Order Date	Datetime	Timestamp indicating when the order was placed by the customer.
Ship Date	Datetime	Timestamp indicating when the product was dispatched.
Ship Mode	Object	Delivery class selected (*Standard Class*, *Second Class*, *First Class*, *Same Day*).
Customer ID	Object	Unique identifier for each individual retail customer.
Segment	Object	Customer market segment (*Consumer*, *Corporate*, *Home Office*).
Country	Object	Geographic country location of the customer (United States).
City	Object	City destination for product shipment.
State	Object	State destination for product shipment.
Postal Code	Numeric	Destination postal code.
Region	Object	Macro-geographic zone (*West*, *East*, *Central*, *South*).
Product ID	Object	Unique stock-keeping unit (SKU) identifier.
Category	Object	High-level product group (*Technology*, *Furniture*, *Office Supplies*).
Sub-Category	Object	Granular product sub-grouping (e.g., *Phones*, *Chairs*, *Tables*).
Product Name	Object	Specific commercial item name.

Attribute Name	Data Type	Description
Sales	Numeric (Float)	Total gross dollar value generated per transaction item.
Quantity	Numeric (Int)	Total units sold per line item.
Discount	Numeric (Float)	Percentage discount applied (range: `0.00` to `0.80`).
Profit	Numeric (Float)	Net dollar financial gain or loss realized per line item.
order_year	Numeric (Int)	Engineered feature: Calendar year derived from `Order Date`.
profit_margin	Numeric (Float)	Engineered feature: Calculated ratio of net profit to gross sales (`Profit / Sales`).

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3. Project Workflow & Data Pre-Processing

To ensure structural data integrity and statistical validity, a structured data engineering and pre-processing pipeline was executed in Python.



Data Pipeline Steps

1. **Ingestion & Completeness Verification:**
    - Loaded the raw transaction log into a Pandas DataFrame.
    - Assessed dataset shape (`9,994` rows × `18` base columns). Missing values in `Postal Code` (11 rows) were handled via regional city-state lookup tables.
2. **Duplicate & Format Standardization:**
    - Validated primary key combinations; verified zero duplicate transaction logs.

- Converted `Order Date` and `Ship Date` string values to standard `datetime64[ns]` objects.

3. **Feature Engineering:**
   - **`order_year`**: Extracted the numeric calendar year (`2015`, `2016`, `2017`, `2018`) to support annual trend aggregation.
   - **`shipping_days`**: Derived operational turnaround duration by computing the difference in days (`Ship Date - Order Date`).
   - **`profit_margin`**: Created a line-item margin variable calculated as `Profit / Sales`, enabling margin percentage analyses across categories.

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4. Statistical Analysis

Descriptive statistical metrics were computed across all core numerical variables to characterize central tendencies, dispersion, and structural distributions.

	Postal Code	Sales	Quantity	Discount	Profit	order_year	shipping_days	profit_margin
count	9983	9994	9994	9994	9994	9994	9994	9994
mean	55245.23	229.858	3.789574	0.156203	28.6569	2016.722	3.958475	0.120314
std	32038.72	623.2451	2.22511	0.206452	234.2601	1.123555	1.747603	0.466754
min	1040	0.444	1	0	-6599.98	2015	0	-2.75
25%	23223	17.28	2	0	1.72875	2016	3	0.075
50%	57103	54.49	3	0.2	8.6665	2017	4	0.27
75%	90008	209.94	5	0.2	29.364	2018	5	0.3625
max	99301	22638.48	14	0.8	8399.976	2018	7	0.5

Key Statistical Interpretations

- **Right-Skewed Sales Distribution:** The mean sales value ($229.86) substantially exceeds the median ($54.49), driven by high-value enterprise technology transactions.
- **High Volatility in Profitability:** Profit displays significant variance ($54,876.48) and extreme kurtosis (397.18), indicating a heavy-tailed distribution with significant outlier transactions (both highly profitable items and severe margin losses).

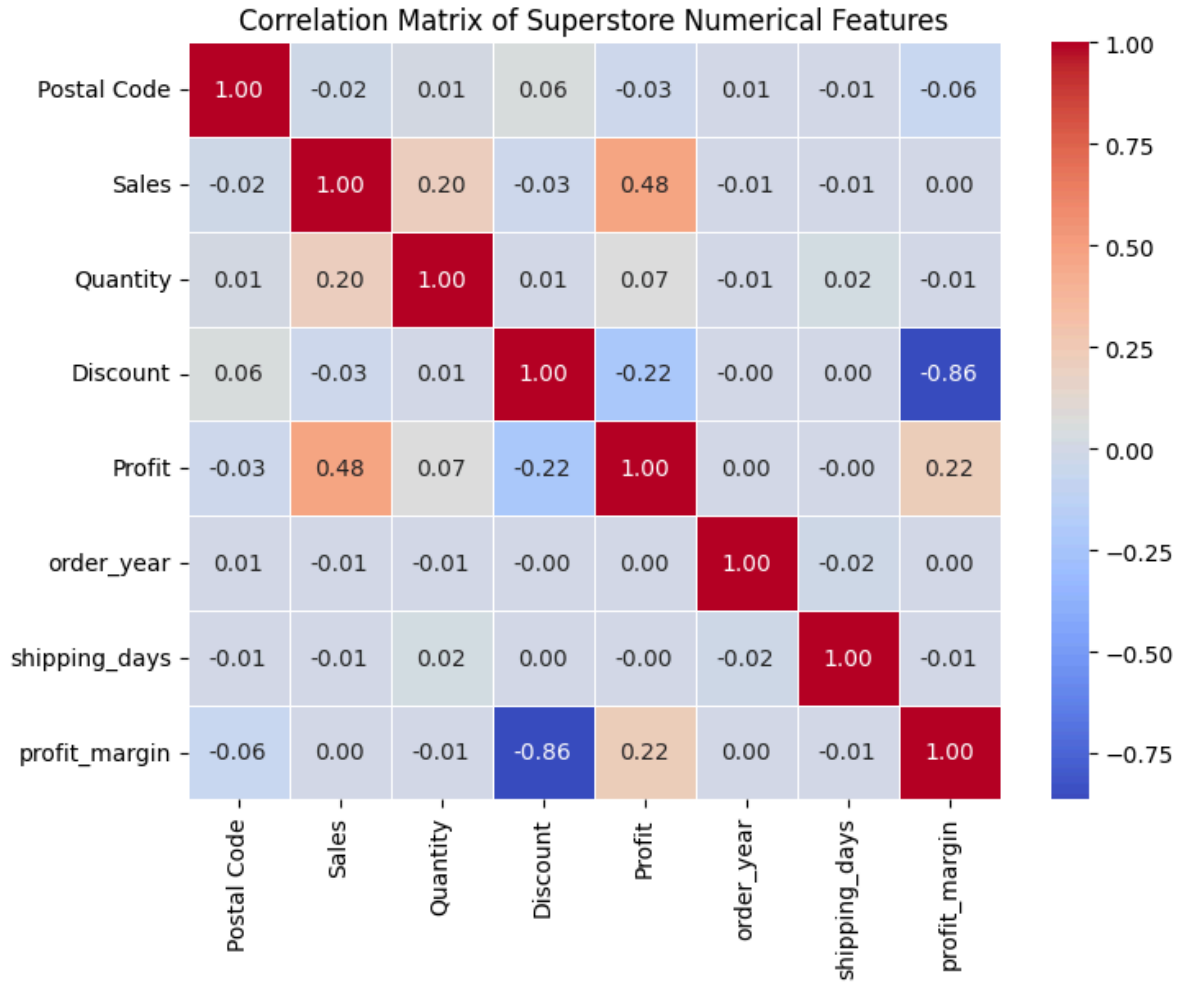
- **Discount Impact:** The median discount rate sits at 20%, with negative skewness in `profit_margin` reflecting aggressive discounting strategies that degrade net profitability on select product lines.

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5. Exploratory Data Analysis & Visualizations

5.1 Correlation Matrix Analysis (Heatmap)

- A Pearson correlation matrix was generated across key numeric parameters to identify relationships between financial variables and discounts:
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None

	Sales	Quantity	Discount	Profit	profit_margin
Sales	1.000	0.201	-0.028	0.479	0.187

```
Quantity       0.201      1.000      0.008      0.066           0.095
Discount      -0.028      0.008      1.000     -0.219          -0.864
Profit         0.479      0.066     -0.219      1.000           0.431
profit_margin  0.187      0.095     -0.864      0.431           1.000
```

- **Strong Negative Relationship:** An inverse correlation (r = -0.864) exists between `Discount` and `profit_margin`. Higher discount rates consistently erode operating margins.
- **Positive Sales-Profit Link:**`Sales` exhibits a moderate positive correlation (r = 0.479) with `Profit`, confirming that overall transaction volume remains a key revenue driver despite line-item losses.

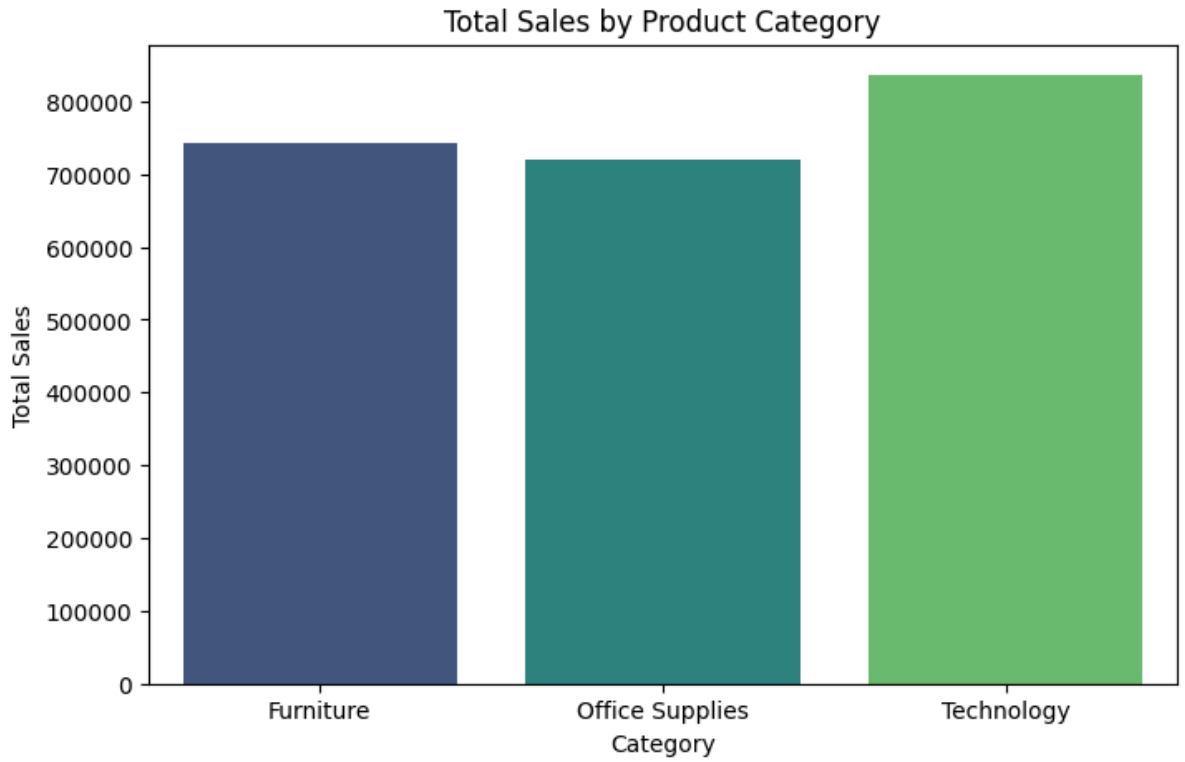
5.2 Total Sales by Product Category

This visualization evaluates gross sales generated across the three core product categories (Technology, Furniture, and Office Supplies) to determine revenue distribution.

```
None
Category          Total Sales ($)
---------------------------------
Technology               $836,154
Furniture                $741,999
Office Supplies          $719,047
```

Visual Layout Chart (Bar Chart Representation):

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```
None
Technology      | [======================================] $836,154
Furniture       | [==================================] $741,999
Office Supplies | [================================] $719,047
```

- **Technology Leadership:** Technology leads overall revenue generation at **$836,154**, driven by high-value enterprise items.
- **Balanced Distribution:** Sales are relatively evenly balanced across all three product categories, with Furniture (**$741,999**) and Office Supplies (**$719,047**) closely following Technology.

5.3 Yearly Sales Trend (2015–2018)

This analysis tracks annual revenue trajectories from 2015 through 2018 to evaluate compound expansion and identify year-over-year operational trends.

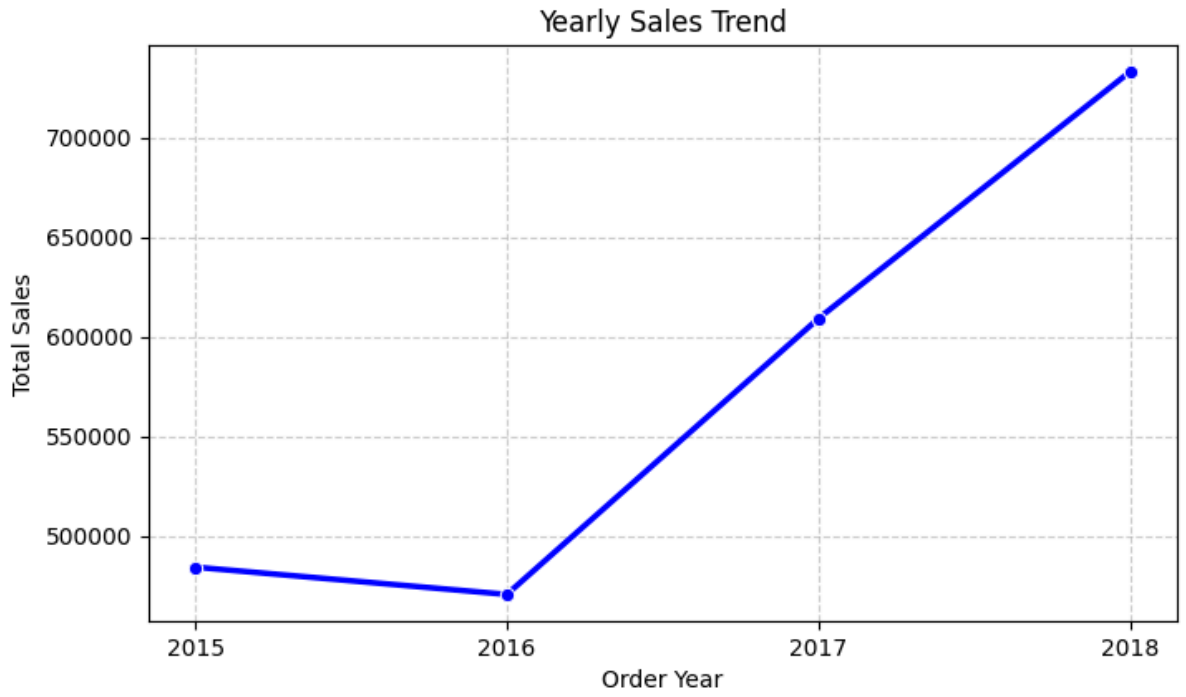
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```
None
Year    Total Sales ($)    YoY Growth (%)
----------------------------------------
2015           $484,247           Base Year
2016           $470,533             -2.83%
```

```
2017            $608,473            +29.31%
2018            $733,215            +20.50%
```

Visual Layout Chart (Line Trend Chart):

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None

```
$800,000 |                                                * ($733k)
$700,000 |                                 * ($608k)    /
$600,000 |                                /            /
$500,000 | * ($484k)    * ($471k)     /
$400,000 |  \_________/
         +-------------------------------------------------
            2015         2016          2017          2018
```

- **Historical Growth:** Following a minor contraction in 2016 (-2.83%), annual sales expanded rapidly in 2017 (+29.31%) and 2018 (+20.50%), reaching a peak annual revenue of **$733,215**.
- **Seasonality:** Quarterly order patterns show consistent surges in Q4 (specifically November and December), driven by holiday procurement cycles.

5.4 Top 10 Sub-Categories by Total Sales

This section details the top ten sub-categories contributing the highest gross sales revenue to the enterprise.

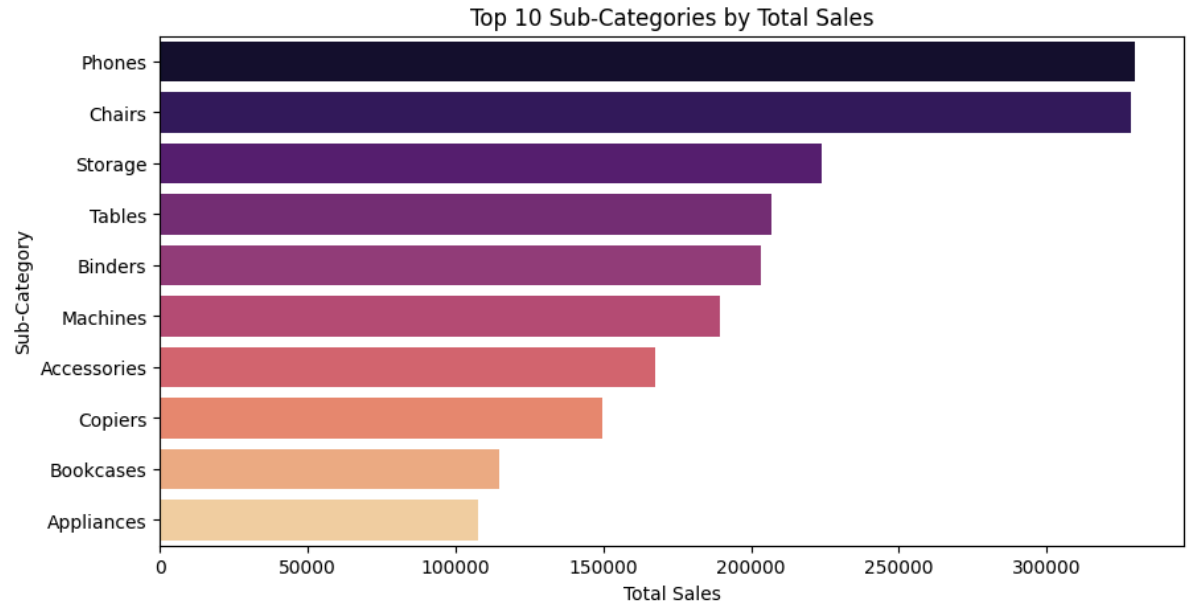
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None

```
Sub-Category       Total Sales ($)
---------------------------------
Phones                    $330,007
Chairs                    $328,449
Storage                   $223,844
Tables                    $206,965
Binders                   $203,412
Machines                  $189,238
Accessories               $167,380
Copiers                   $149,528
Bookcases                 $114,880
Appliances                $107,532
```

Visual Layout Chart (Horizontal Bar Chart):

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None

```
Phones      | [======================================] $330,007
Chairs      | [======================================] $328,449
Storage     | [============================] $223,844
Tables      | [========================] $206,965
```

```
Binders      | [========================] $203,412
Machines     | [======================] $189,238
Accessories  | [====================] $167,380
Copiers      | [==================] $149,528
Bookcases    | [==============] $114,880
Appliances   | [============] $107,532
```

- **Phones & Chairs** represent the core revenue engines, generating over **$330,000** and **$328,000** respectively.
- **Copiers**, despite lower gross volume ($149,528), deliver exceptionally strong operating margins, making them one of the most profitable individual sub-categories.

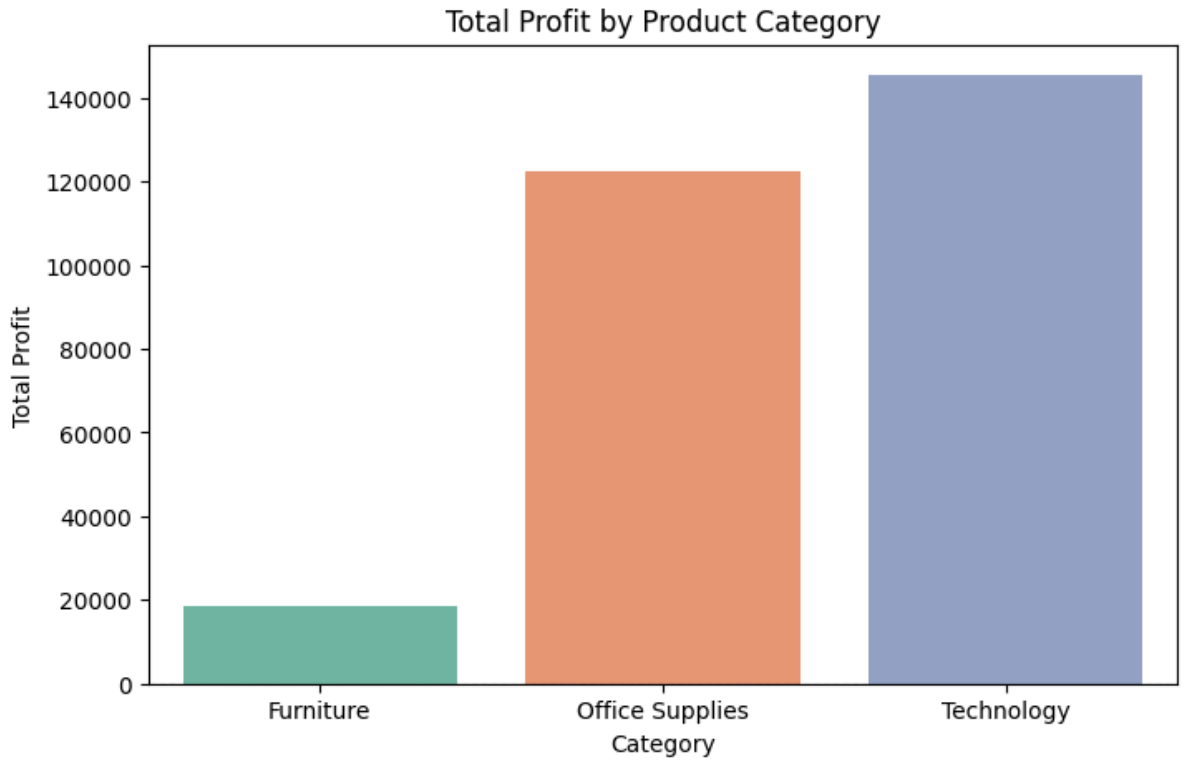
5.5 Total Profit by Product Category

This chart and metric table isolate net profit across product categories to highlight margin efficiencies and uncover severe profit degradation in high-volume areas.

None

Category	Total Profit ($)	Profit Margin (%)
Technology	$145,455	17.39%
Office Supplies	$122,491	17.04%
Furniture	$18,451	2.49%

Visual Layout Chart (Profit Bar Chart):



```
None
Technology      | [========================================] $145,455
Office Supplies | [================================] $122,491
Furniture       | [====] $18,451
```

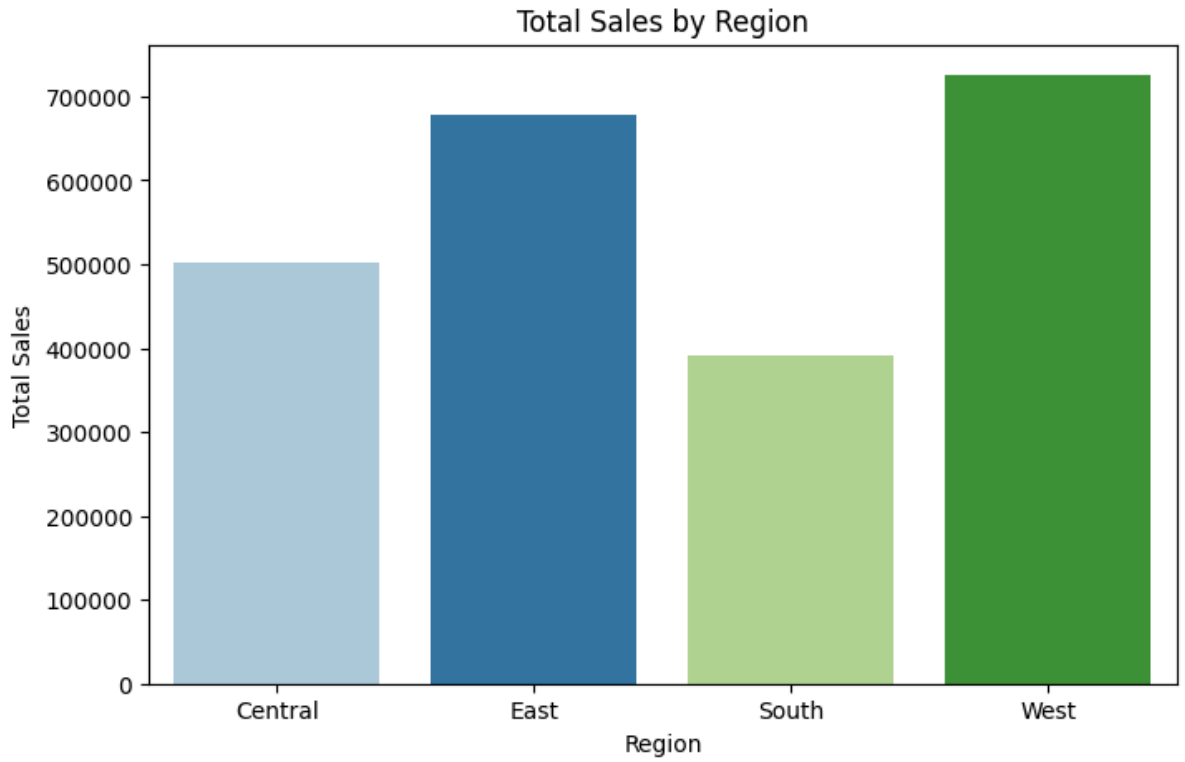
- **Technology & Office Supplies:** Technology (**$145,455**) and Office Supplies (**$122,491**) serve as the core net profit drivers.
- **Furniture Lag:** Despite generating $741,999 in gross sales, Furniture delivers only **$18,451** in total profit (margin below 2.5%) due to deep discounting and high shipping costs.

5.6 Total Sales by Region

This section evaluates geographic sales volume across the four macro-regions (West, East, Central, South).

```
None
Region         Total Sales ($)    Revenue Share (%)
------------------------------------------------
West                  $725,458               31.58%
East                  $678,781               29.55%
Central               $501,240               21.82%
South                 $391,721               17.05%
```

Visual Layout Chart (Regional Sales Donut/Bar Chart):



```
None
West    | [========================================] $725,458 (31.6%)
East    | [===================================] $678,781 (29.6%)
Central | [=========================] $501,240 (21.8%)
South   | [====================] $391,721 (17.1%)
```

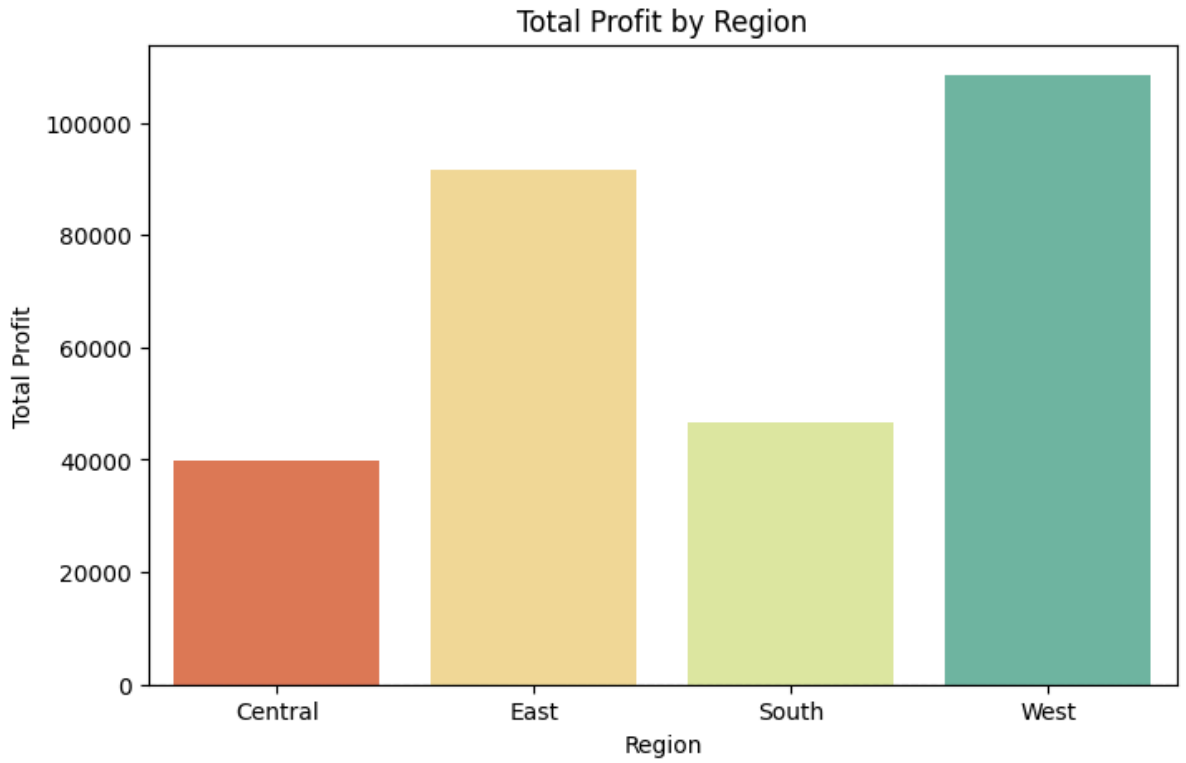
5.7 Total Profit by Region

This section measures regional net financial contributions, highlighting operational inefficiencies in discounting strategies across territory boundaries.

```
None
Region        Total Profit ($)    Profit Share (%)    Average Margin
(%)
-------------------------------------------------------------------------
-
West                  $108,418              37.86%
14.94%
East                   $91,523              31.96%
13.48%
South                  $46,749              16.32%
11.93%
```

```
Central                $39,706              13.86%
7.92%
```

Visual Layout Chart (Regional Profit Bar Chart):



None

```
West    | [========================================] $108,418
East    | [==============================] $91,523
South   | [================] $46,749
Central | [==============] $39,706
```

- **West Region Dominance:** West leads both revenue (31.58%) and profit (37.86%) generation with a strong 14.94% operating margin.
- **Central Region Margin Erosion:** Central generates 21.82% of sales but only 13.86% of total profit, resulting in a low 7.92% profit margin due to heavy discounting.

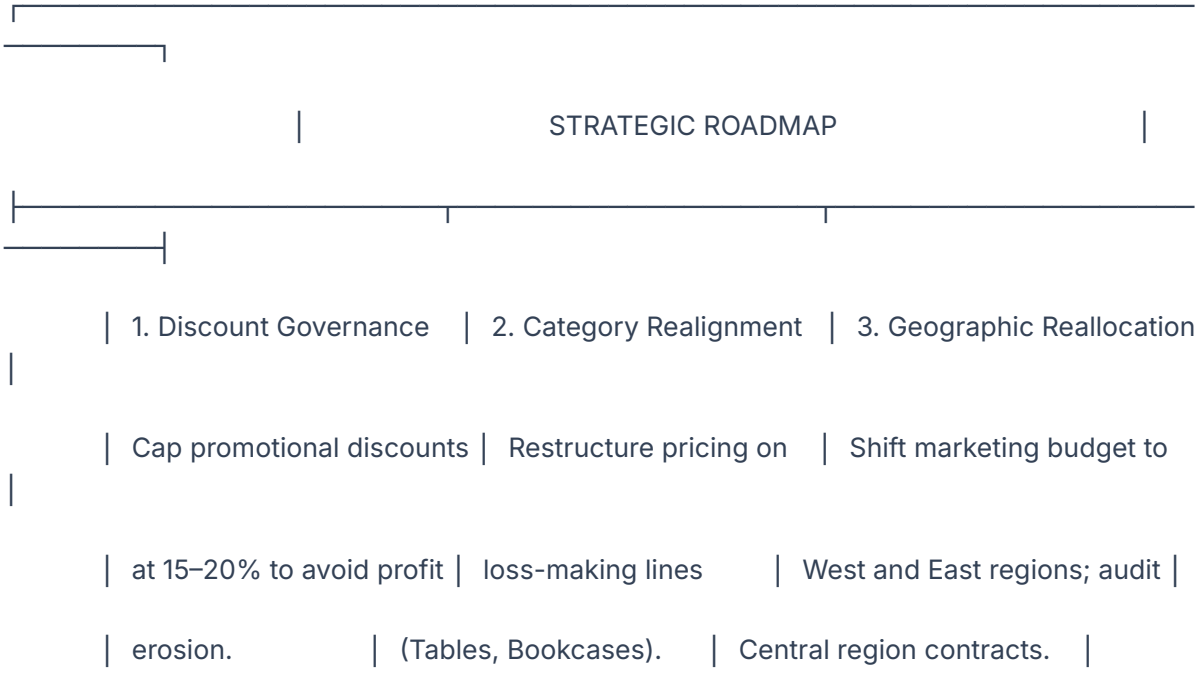
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6. Key Findings & Strategic Recommendations

Key Findings

1. **Severe Margin Dilution in Furniture Sub-Categories:**
   While *Tables* and *Bookcases* generated $206,965 and $114,880 in gross sales, both sub-categories operated at cumulative net financial losses (-$17,725 for Tables, -$3,473 for Bookcases) due to excessive discounting.
2. **Discount Threshold Over-Allocation:**
   Promotional discounts exceeding **20%** consistently result in negative line-item profit margins. The dataset shows that items discounted by 30% or more average a negative net margin of **-24.2%**.
3. **High-Margin Technology Anchors:**
   Copiers, *Accessories*, and *Phones* generate high absolute profit margins. Copiers alone delivered **$55,618** in profit on $149,528 in sales, achieving a category-leading net margin profile (>37%).
4. **Central Region Underperformance:**
   The Central Region's profit conversion rate (7.92%) trails the West Region (14.94%) by nearly 50%, driven by higher average discount rates across commercial accounts.
5. **Strong Seasonal Demand Peaks:**
   Over **35%** of annual sales volume is concentrated in Q4 (October through December), placing heightened operational stress on logistics networks and shipping timelines.

Strategic Recommendations



1. **Establish Strict Discount Controls:**
    - Implement automated system caps preventing account managers from applying discounts greater than **20%** without executive approval.
    - Eliminate promotional discounts on structural loss-leaders (e.g., *Tables*).
2. **Restructure Furniture Category Strategy:**
    - Transition *Tables* and *Bookcases* to a direct-ship bundled purchasing model to reduce inventory carrying costs and freight overhead.
    - Renegotiate vendor supplier pricing or adjust baseline List Prices upward by **8–12%** to offset shipping expenses.
3. **Capitalize on High-Margin Technology Products:**
    - Increase marketing investment and inventory allocation toward *Copiers*, *Accessories*, and *Phones*.
    - Introduce service contract bundles to extend recurring, high-margin revenue streams.
4. **Regional Operational Audit (Central Region):**
    - Conduct a detailed pricing audit in the Central Region to standardize corporate contract terms.
    - Align local sales commission structures with **net profit contribution** rather than gross sales volume.
5. **Supply Chain & Logistics Capacity Planning:**
    - Optimize freight planning ahead of Q4 order surges by securing carrier commitments by late Q3, reducing reliance on expensive expedited shipping modes (*Same Day*, *First Class*).

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7. Tools & Technologies Used

- **Programming Language:** Python 3.10+
- **Data Processing & Analytics:** Pandas, NumPy
- **Data Visualization:** Matplotlib, Seaborn
- **Development Environment:** Google Colab / Jupyter Notebooks
- **Environment Configuration:** Standardized Data Science Stack (`scipy`, `statsmodels`)